

Basic Principles of Medicine 2
Module: Digestive system and Metabolism (DM)
Lecture No: DM 27

**Lecture Title: Metabolic roles of Folic Acid and
vitamin B12**

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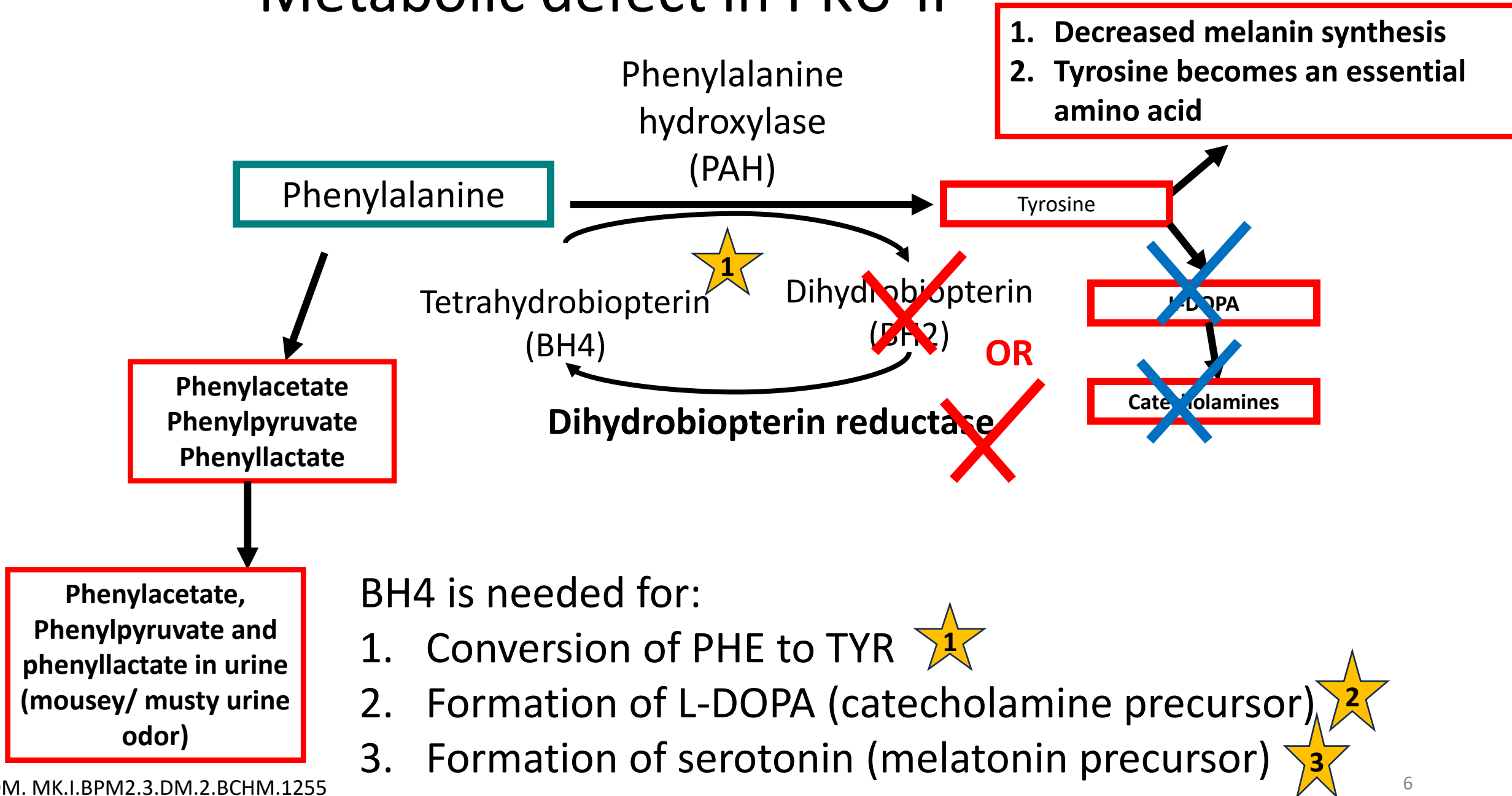
Objectives: Metabolic roles of Folic acid (B9) and vitamin B12

SOM. MK.I.BPM2.3.DM.2.BCHM.1255	Discuss the role of BH4 (tetrahydrobiopterin) in metabolism and effects of tetrahydrobiopterin deficiency (PKU II)
SOM. MK.I.BPM2.3.DM.2.BCHM.1256	Describe the clinical features and laboratory findings associated with folic acid deficiency, B12 deficiency, and inherited homocystinuria
SOM. MK.I.BPM2.3.DM.2.BCHM.1257	Explain mechanism of methyl-folate trap in B12 deficiency
SOM. MK.I.BPM2.3.DM.2.BCHM.1258	Specify one- carbon donors in metabolism and the groups they donate (SAM, THF, Cobalamin)
SOM. MK.I.BPM2.3.DM.2.BCHM.1259	Explain the formation and functions of S-Adenosyl methionine (SAM)
SOM. MK.I.BPM2.3.DM.2.BCHM.1260	Analyze metabolism of homocysteine, vitamins/ coenzymes required for metabolism & interpret clinical significance of homocysteine
SOM. MK.I.BPM2.3.DM.2.BCHM.1261	Explain formation of THF from folate and specify the mechanism of action of their inhibitors
SOM. MK.I.BPM2.3.DM.2.BCHM.1262	Summarize the role of folic acid as an acceptor and donor of 1-carbon groups in metabolism. Explain the role of folic acid as an acceptor of one carbon groups from amino acid metabolism and their utilization for nucleotide synthesis
SOM. MK.I.BPM2.3.DM.2.BCHM.1263	Indicate different forms of THF (formyl, methylene and methyl) and reactions requiring different forms of THF
SOM. MK.I.BPM2.3.DM.2.BCHM.1264	Identify metabolic reactions requiring vitamin B12 as a coenzyme.
SOM. MK.I.BPM2.3.DM.2.BCHM.1265	Compare and contrast causes, clinical and laboratory findings in folate and vitamin B12 deficiency
SOM. MK.I.BPM2.3.DM.2.BCHM.1266	Indicate role of intrinsic factor in vitamin B12 absorption

Recommended reading:

- Lippincott's Biochemistry 9th Edition; Chapter: Micronutrients: Vitamins
 <https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences.lwwhealthlibrary.com/book.aspx?bookid=3402>
- Lippincott's Biochemistry 9th Edition: [Micronutrients: Vitamins | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#) Sections on Vitamin B9 and vitamin B12; Figure 28.29: Summary of Vitamins
 - Questions 28.6

Metabolic defect in PKU-II



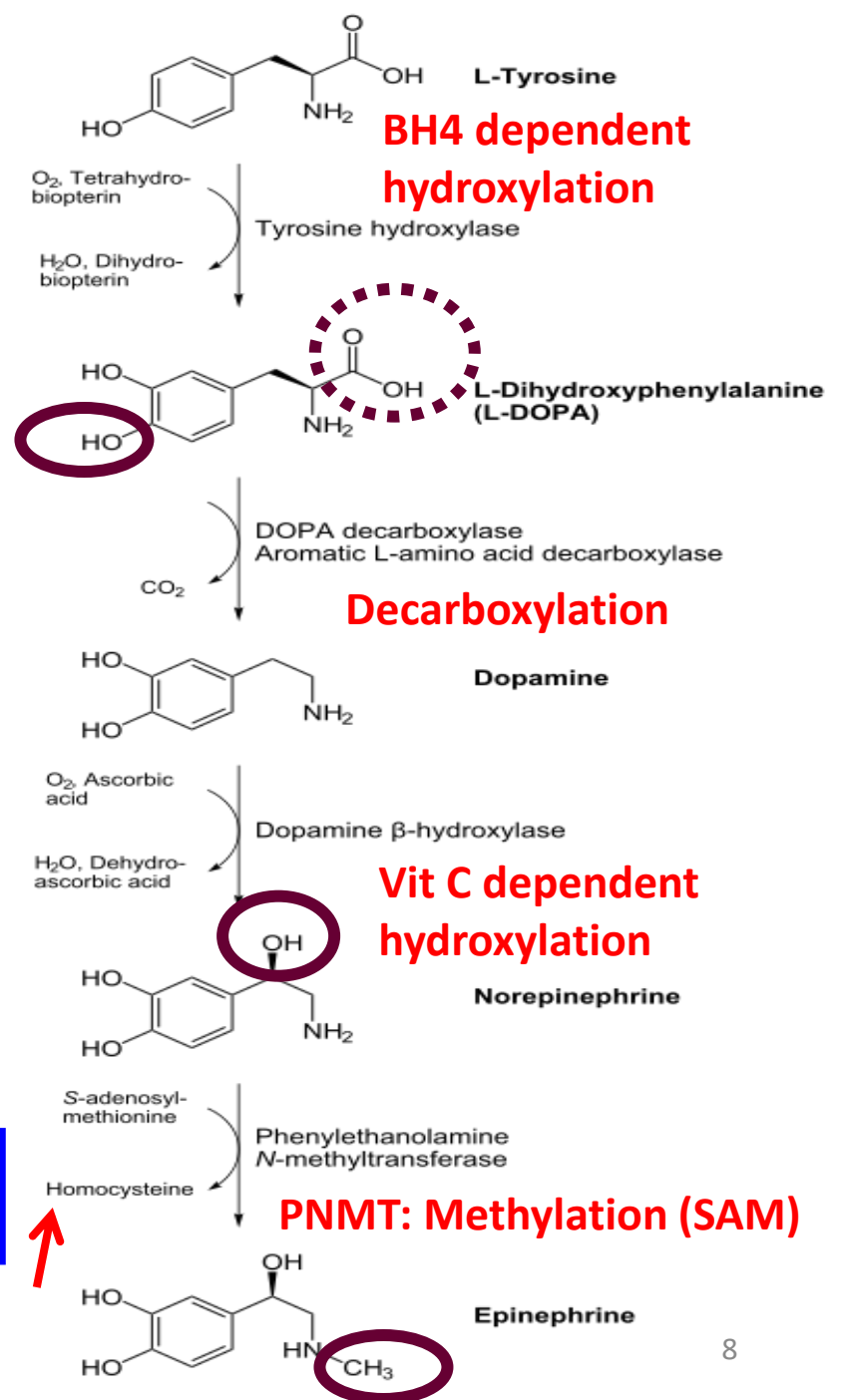
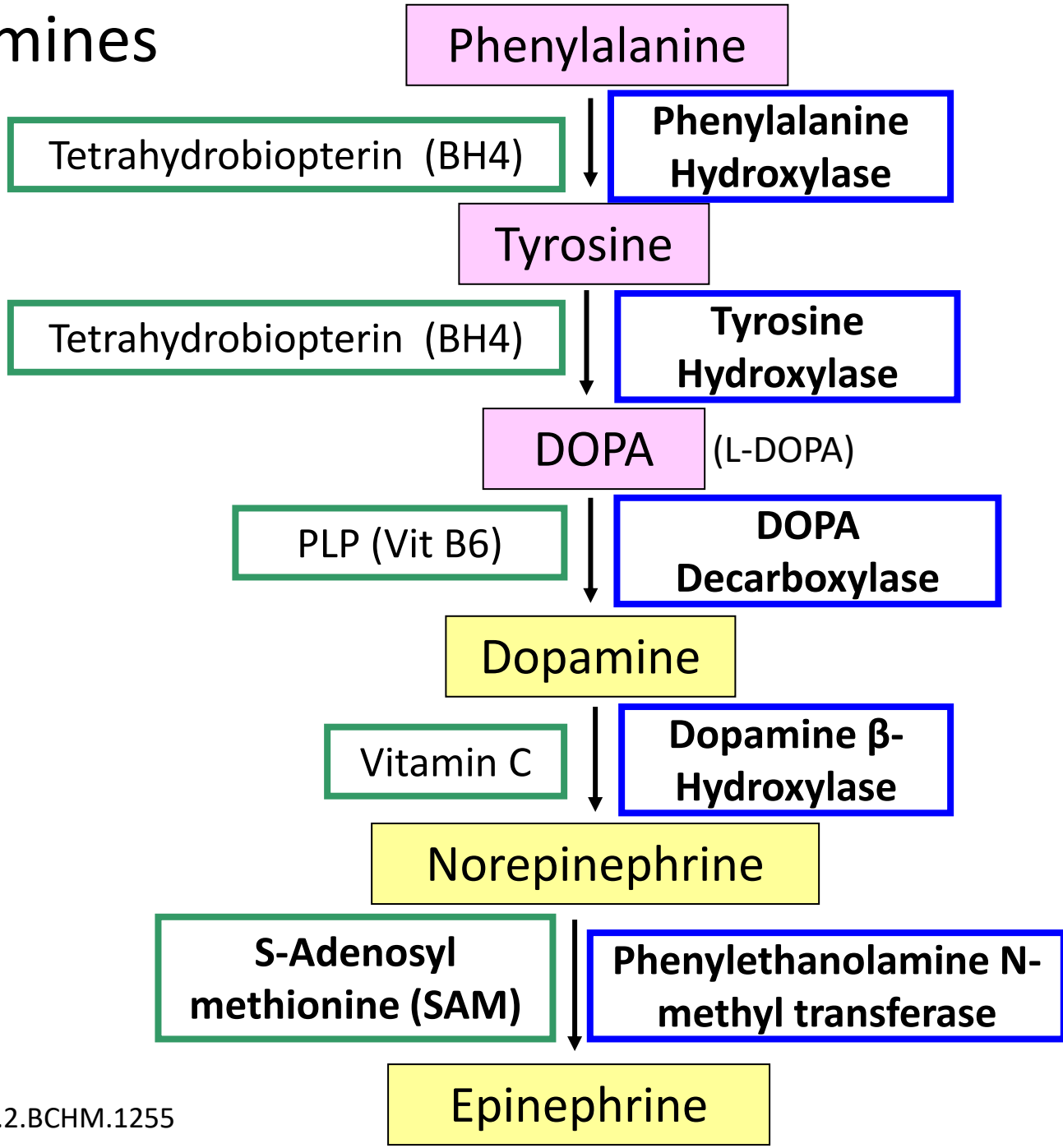
PKU-II (Malignant PKU)

- Deficiency of dihydrobiopterin synthesis or dihydrobiopterin reductase (BH2/BH4)
- Much more severe CNS symptoms
 - Decreased neurotransmitter synthesis- serotonin, and the catecholamines: (dopamine, norepinephrine, epinephrine)
- Treatment includes dietary Phe restriction and providing dietary biopterin and precursors of the neurotransmitters
 - Treatment with **Saproterin** (structural analogue of BH4)
 - Difficulties remain though because of Blood Brain Barrier

Catecholamines

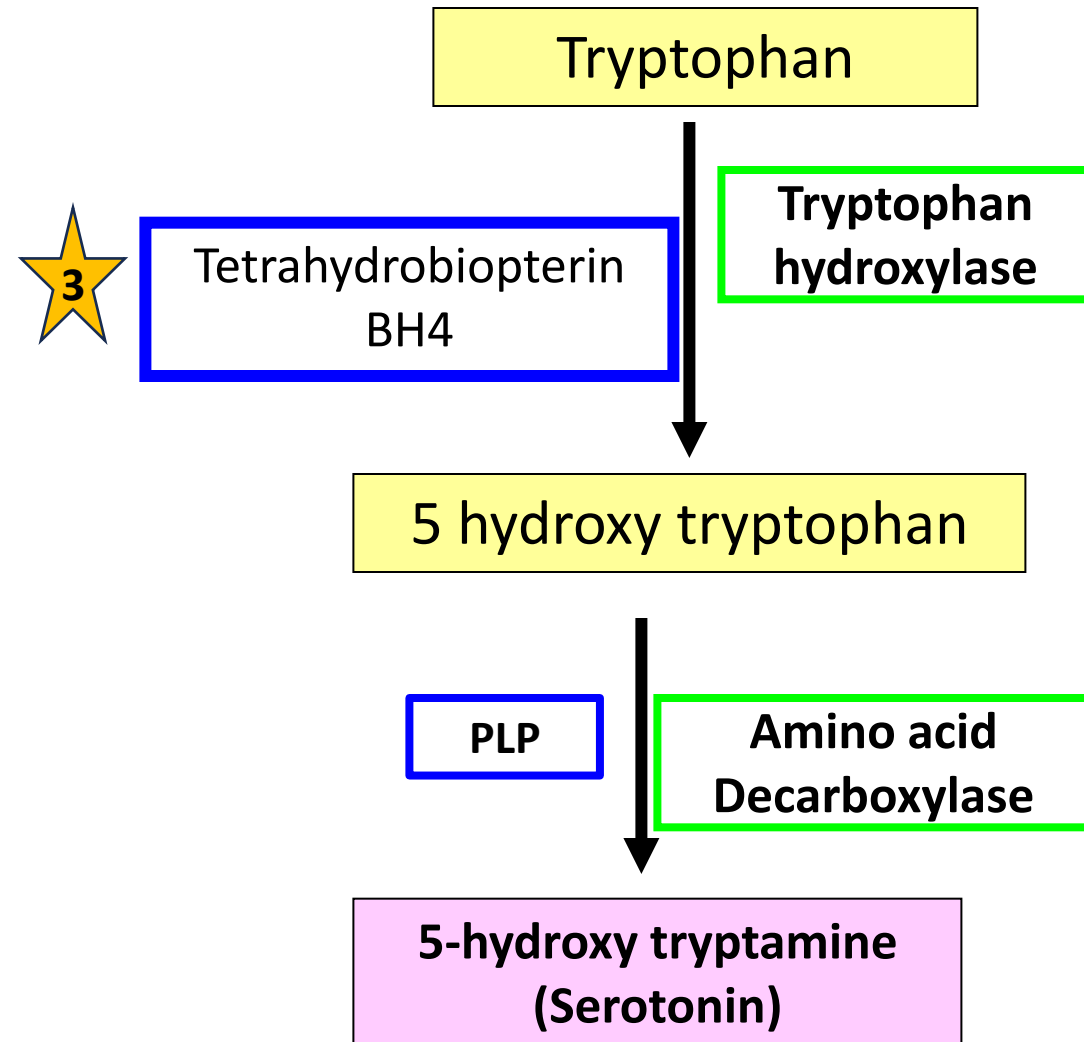
★ 1

★ 2



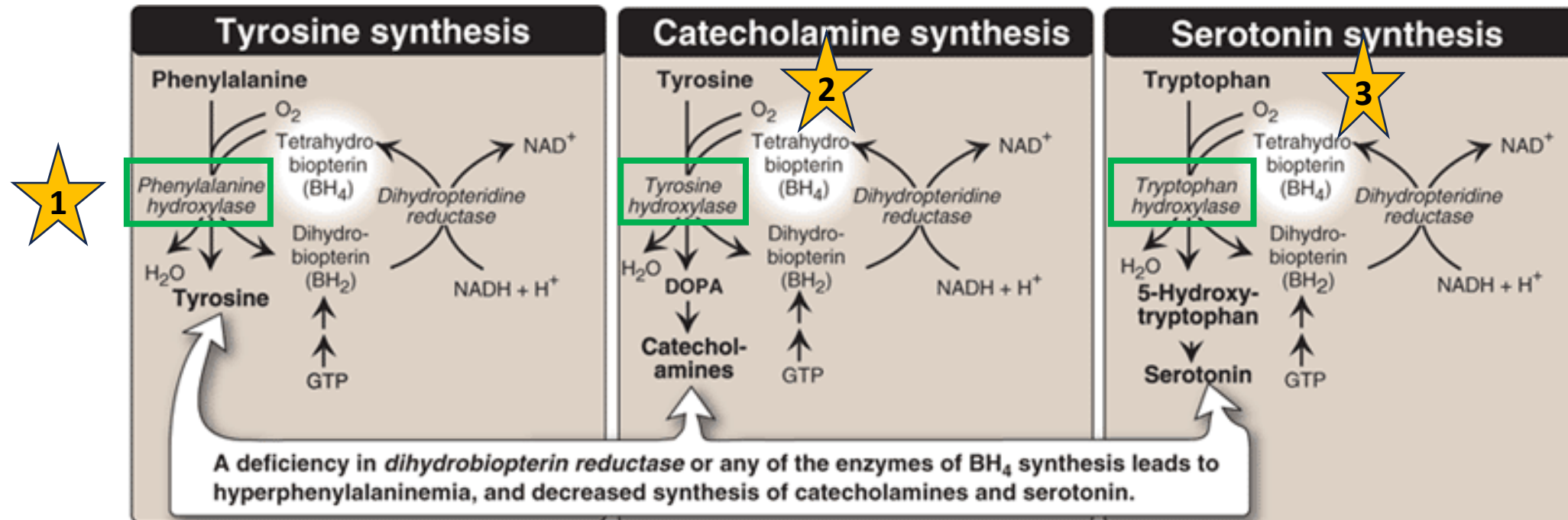
Serotonin (5-hydroxy tryptamine) synthesis

- Serotonin is synthesized in the gut, platelets & CNS
- Precursor: Tryptophan
- Serotonin metabolized to 5-hydroxyindole acetic acid (5-HIAA) by MAO



PKU-II (Malignant PKU)

- Deficiency of dihydrobiopterin synthesis **or** dihydrobiopterin reductase results in hyperphenylalaninemia & decreased synthesis of neurotransmitters (catecholamines, serotonin).
 - Results in intellectual disability and seizures
- Management includes addition of dietary tetrahydrobiopterin
 - Provision of neurotransmitters precursors in the diet
 - Dietary Phe restriction



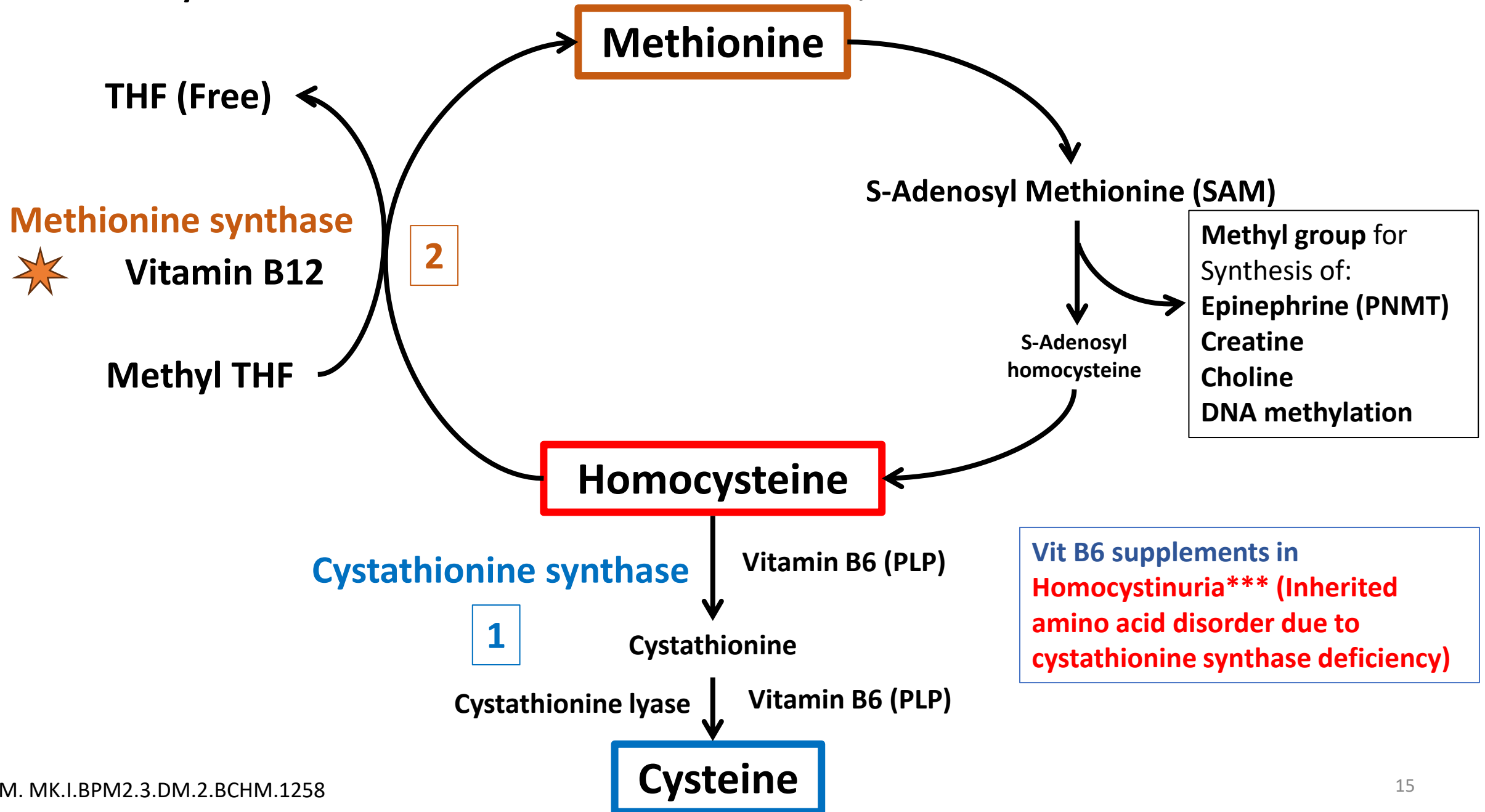
Phenylketonuria: PKU-I and II

- Elevated phenyllactate, and phenylpyruvate levels (mousey/musty urine)
- Can be due to deficiency of PAH enzyme (PKU-I; Autosomal recessive)
- Can be due to deficiency of BH4 (PKU-II)
 - Due to inability to synthesize BH4
 - Or due to inability to reduce BH2 to BH4

Role of folate (vitamin B9) and vitamin B12

- Metabolic roles of folic acid and vitamin B12 are closely inter-related
- **Folate and vitamin B12 deficiency affects rapidly dividing cells (RBC precursors) → Macrocytic megaloblastic anemia**
- Differentiate between folate and B12 deficiency
- **Macrocytic megaloblastic anemia: Treat with B12 and folate**
- Both are required for **homocysteine metabolism**

Homocysteine metabolism: Role of PLP; Vitamin B12 and Folate



Methionine and S-Adenosyl Methionine (SAM)

- SAM synthesized from Methionine
- **SAM is Methyl group donor** in
 - **Epinephrine** Synthesis (PNMT)
 - **Creatine** Synthesis
 - **Choline** Formation (phosphatidylcholine, and acetylcholine)
 - **Methylation** of DNA and histone (*recall genetics; epigenetic gene regulation*)
- After donating methyl group, SAM is converted to **homocysteine**
- **Homocysteine** can have **two fates**:
 1. Converted to **Cysteine** (PLP) (**Cystathionine synthase**)
 2. Converted back to **Methionine** (B₁₂ and methyl-tetrahydrofolate) (**Methionine synthase**)

Vitamin B9 (Folate/ folic acid)

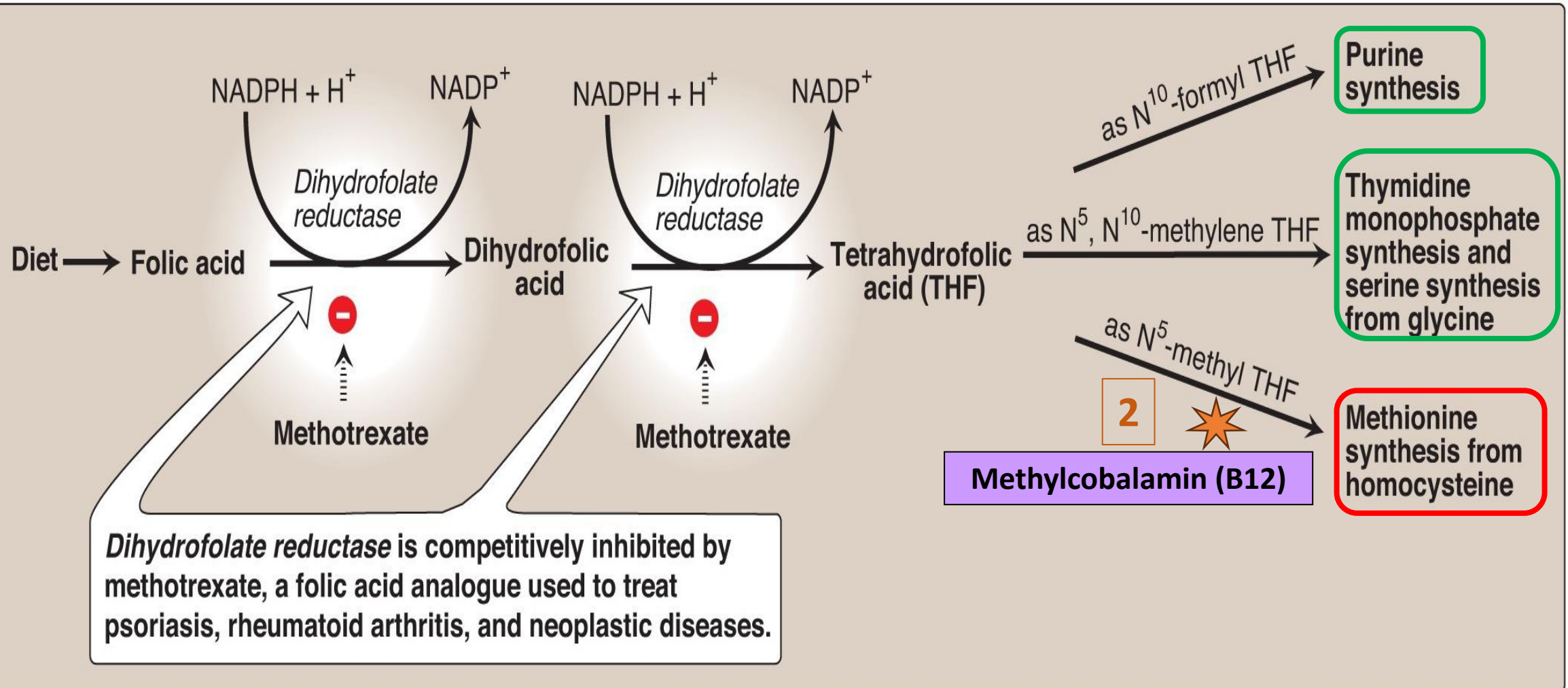
- Green leafy vegetables (rich sources)
- Folate deficiency during pregnancy (preconception & first month of conception) associated with high risk of fetal **neural tube defects** (spina bifida/anencephaly)
- Folate deficiency (high risk groups)
 - Pregnancy (increased requirement)
 - Chronic alcoholics
 - Persons who avoid vegetables in the diet
 - Use of dihydrofolate reductase inhibitors (methotrexate)
- Fortification of staple food



**Folate named from
“*foliage*” – Green leafy
plants**

Metabolic role of folate

Purine & pyrimidine
(thymidine) synthesis

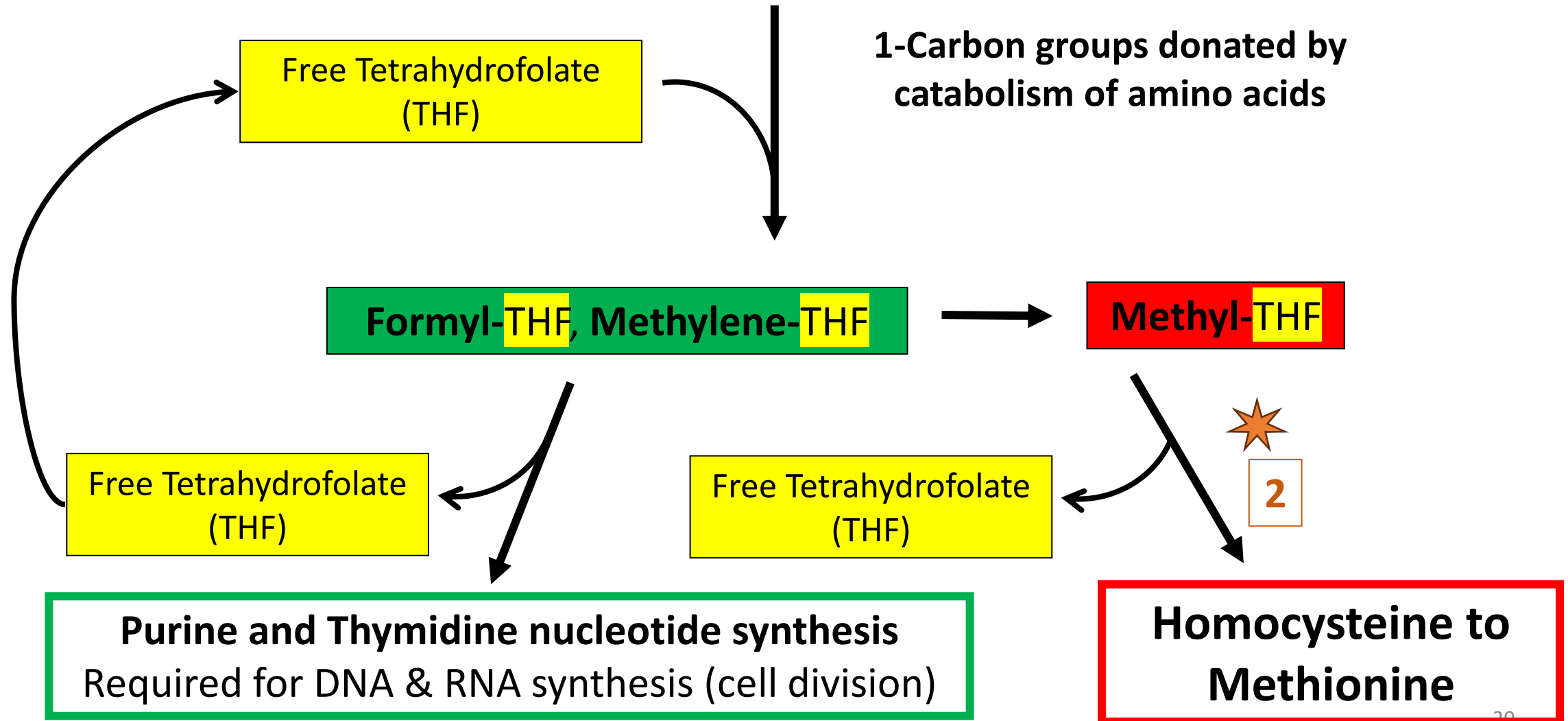


Metabolic role of folate

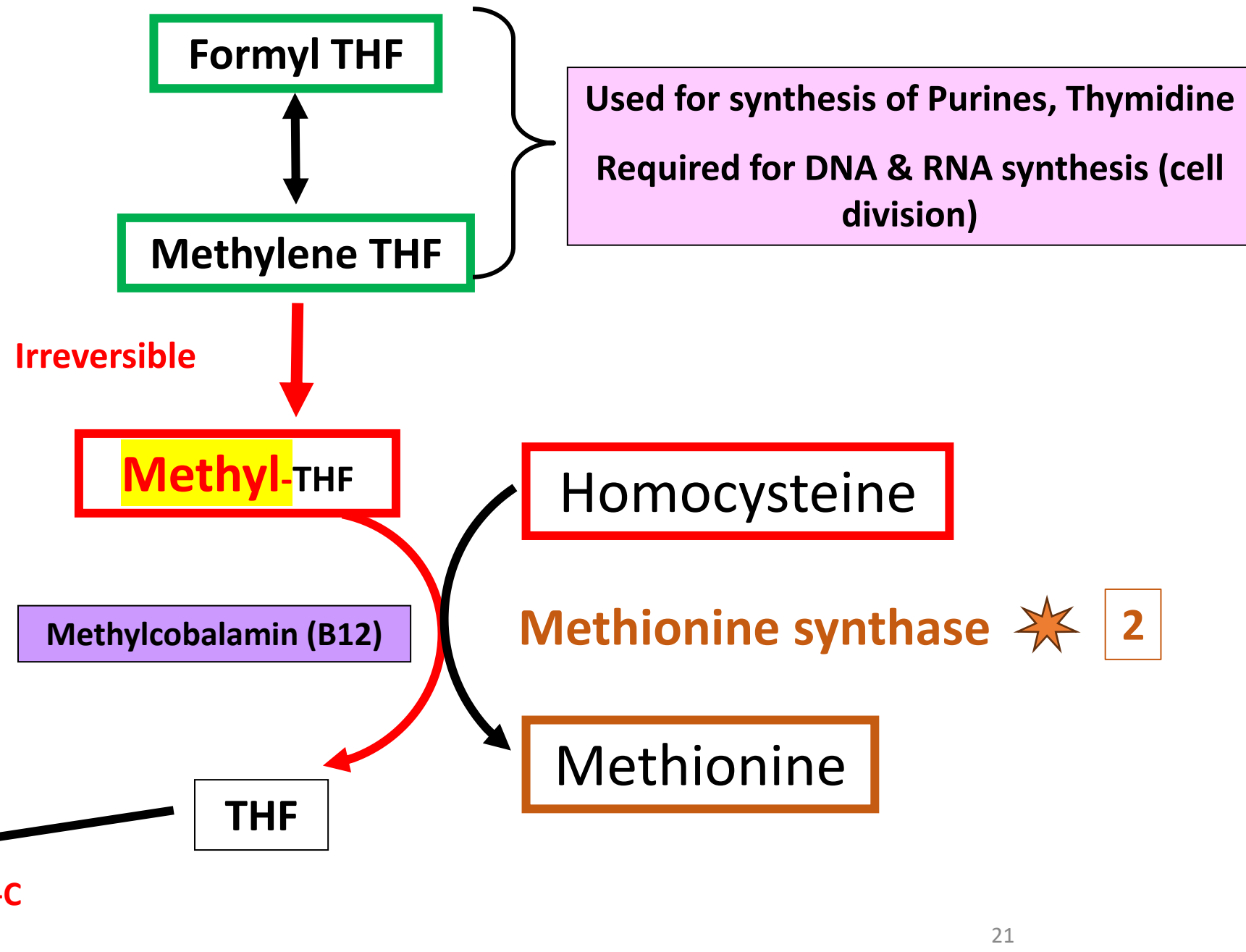
- Coenzyme form is **tetrahydrofolate**
 - Acts as a carrier of 1-C groups (**methyl**, **formyl**, **methylene**)
- **Formyl THF and methylene THF:**
 - Donates C-atoms for **Purines & pyrimidine (thymidine)** nucleotide synthesis
 - Required by rapidly dividing cells for **DNA synthesis**
- **Methyl** THF used ONLY in one reaction:
 - Conversion of **Homocysteine to Methionine**
 - Required for **methionine synthase** (Homocysteine metabolism)
 - Remember this reaction also requires B₁₂ (cobalamin)
 - 'STORAGE' form of THF
 - NOT used for purine and pyrimidine synthesis

Tetrahydrofolate (THF): Reversible carrier of 1-C groups

Amino acid metabolism forms 1-C groups (Glycine, Serine..)



Metabolic role of folate (vitamin B9)



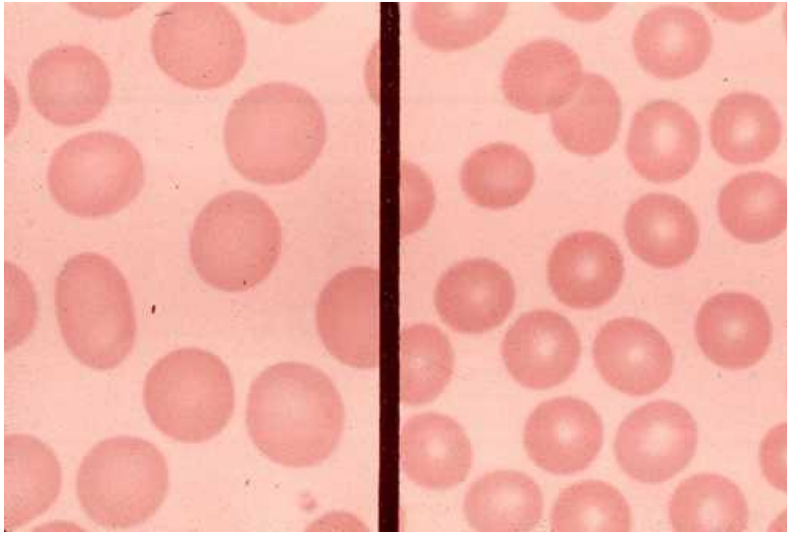
Effects of Folate deficiency

- Reduction in 1-C donors (methylene and formyl THF) → **Reduced Purine & pyrimidine nucleotide synthesis** → Delay in DNA synthesis resulting in reduced rates of cell division (Mitotic lag)
 - Cytosol enlarges but delay in DNA synthesis (mitotic lag)
 - Affects RBC formation (erythropoiesis) – **Macrocytes and Megaloblasts** observed in blood smears
 - Decrease in hemoglobin concentration (**macrocytic, megaloblastic anemia**)
 - WBC: Hypersegmented neutrophils in peripheral blood smear
- **Elevated Homocysteine levels; Methylmalonate levels are normal**

Folic acid and vitamin B12 deficiency

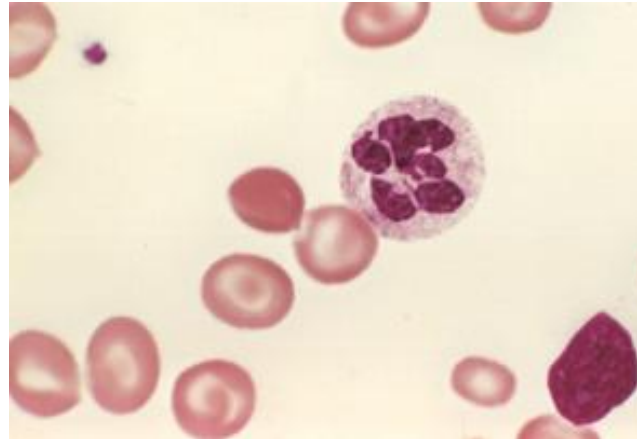
Macrocytic

Normal

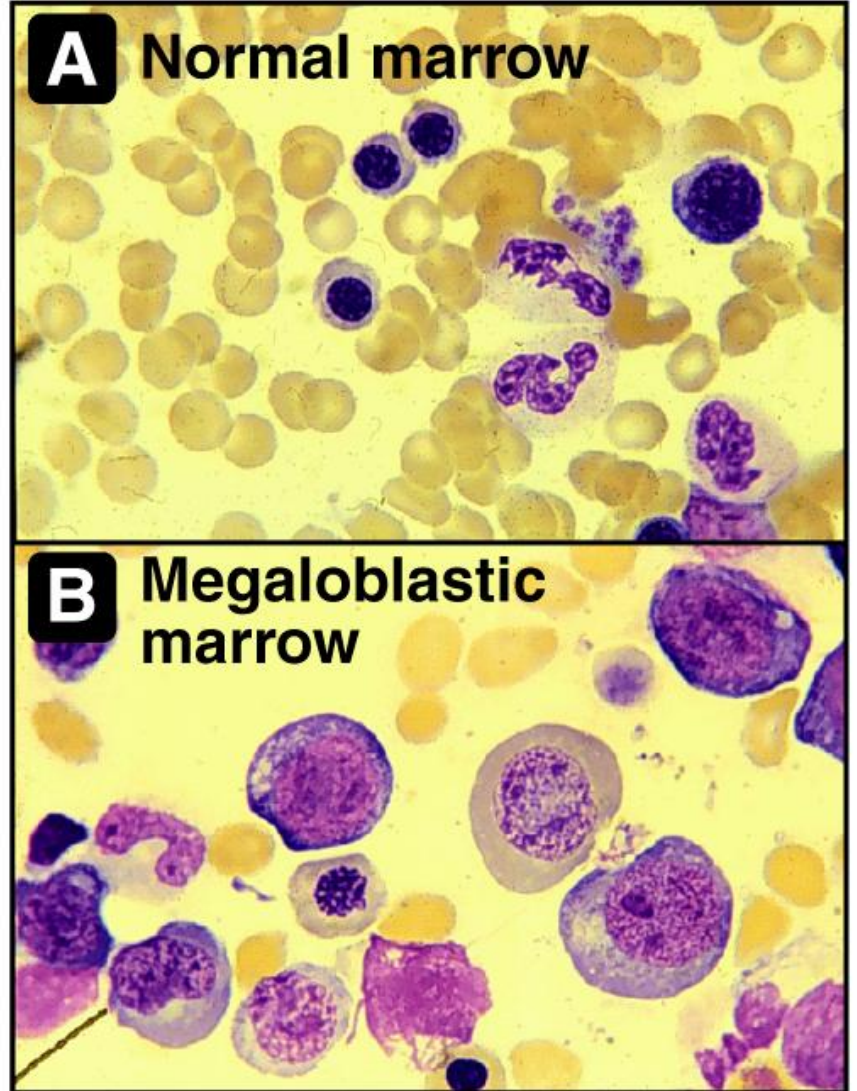


>100 fL mean corpuscular volume (MCV); Large average RBC size

Normal MCV = 80-100 fL



Hypersegmented neutrophils (>3-5 lobes)



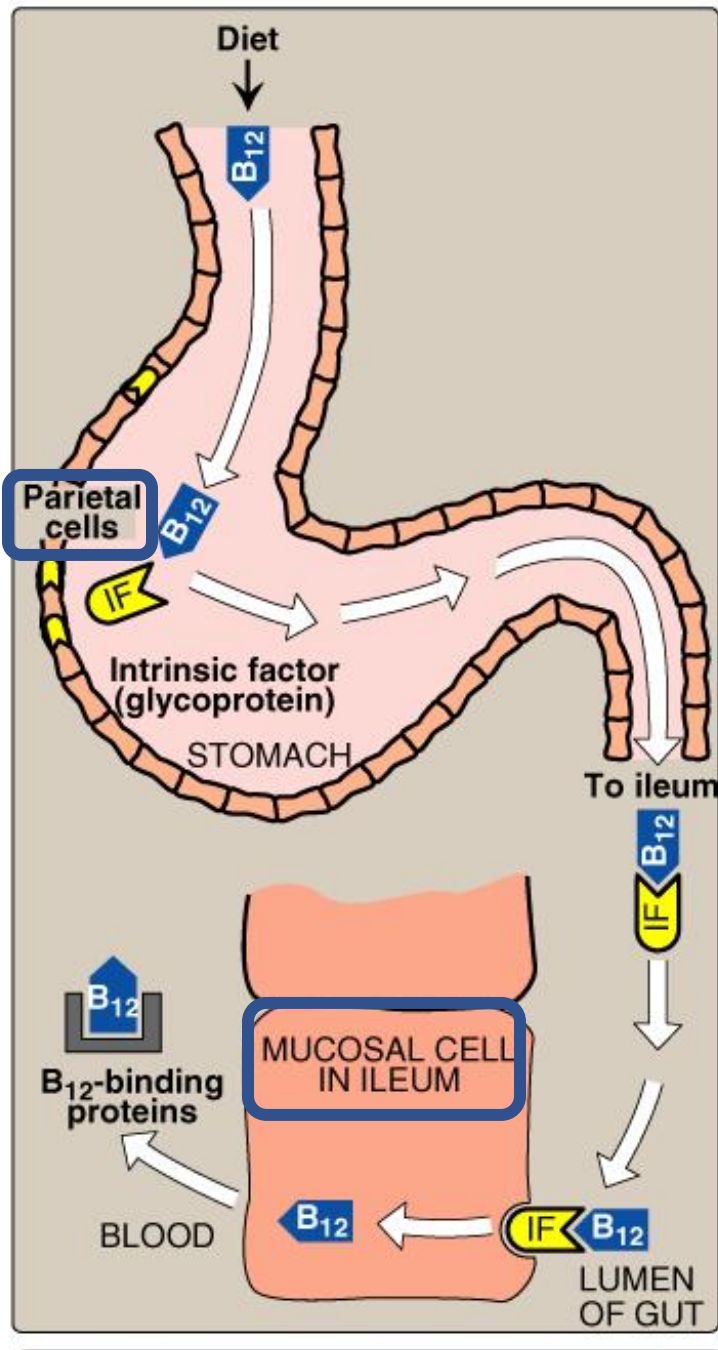
Vitamin B₁₂ (Cobalamin): Absorption



- Dietary sources: Animal products
- Vitamin B₁₂ is stored in liver, deficiency usually takes some time to develop
- **Vitamin B12 absorption:**
 - **Intrinsic factor (IF)** released from **Parietal cells of stomach** (recall parietal cells also secrete HCl).
Intrinsic factor binds to vitamin B12.
 - Vitamin B₁₂ – IF complex absorbed in **terminal ileum**
- Vitamin B₁₂ deficiency: **Macrocytic anemia** and **neurological features** (tingling sensation in the hands and feet, inability to walk, psychiatric disturbances, urinary incontinence.....)

Risk factors for B12 deficiency

- **Vegans**
- Failure to absorb B₁₂ due to
 - Atrophic gastritis and **achlorhydria** (Parietal cell loss)
 - Bariatric surgery
 - Pernicious anemia (autoimmune destruction of Parietal cells)
 - Disease of the **Ileum** (tuberculosis, Crohn's disease)

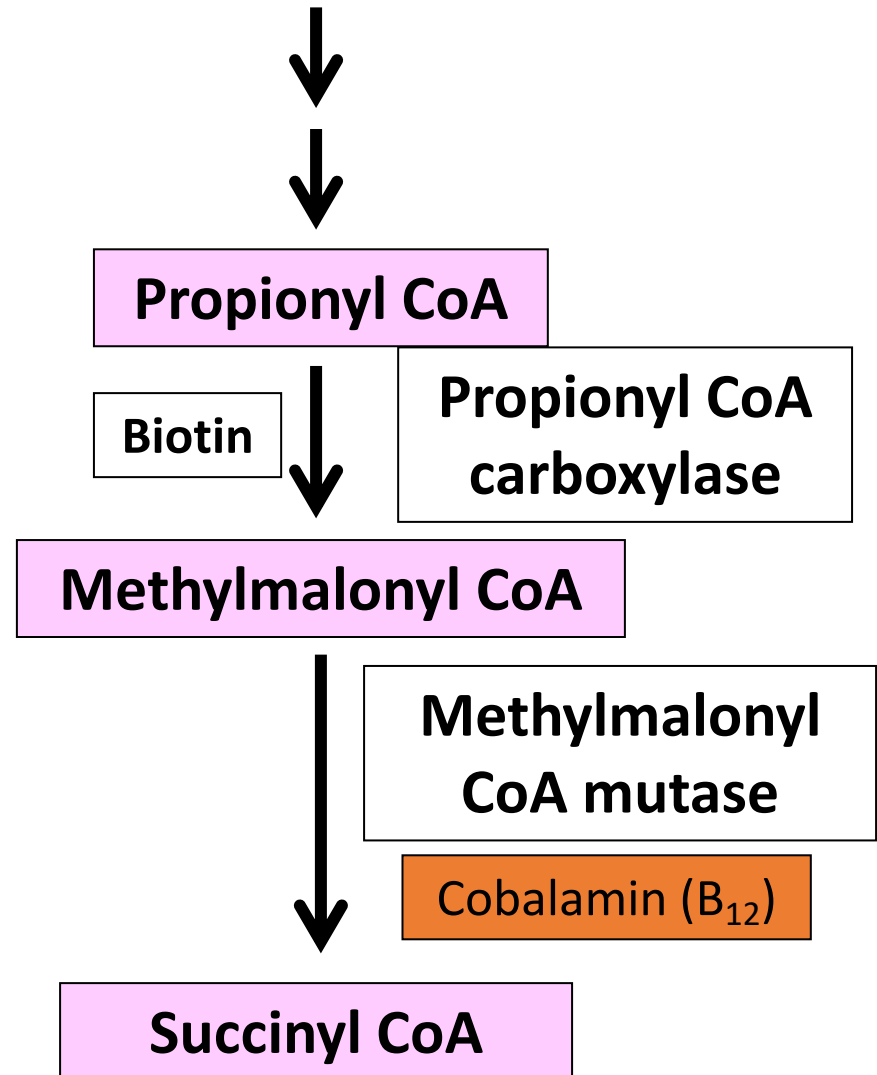


Reactions requiring vitamin B12:

1. Methylmalonyl CoA mutase

- Serum methylmalonate levels are elevated in vitamin B12 deficiency (not in folate deficiency)
- **Neurological complications** due to demyelination due to accumulation of **methylmalonate**
- Vit B12 supplements in methylmalonic aciduria***
(Inherited amino acid disorder due to methylmalonyl CoA mutase deficiency)

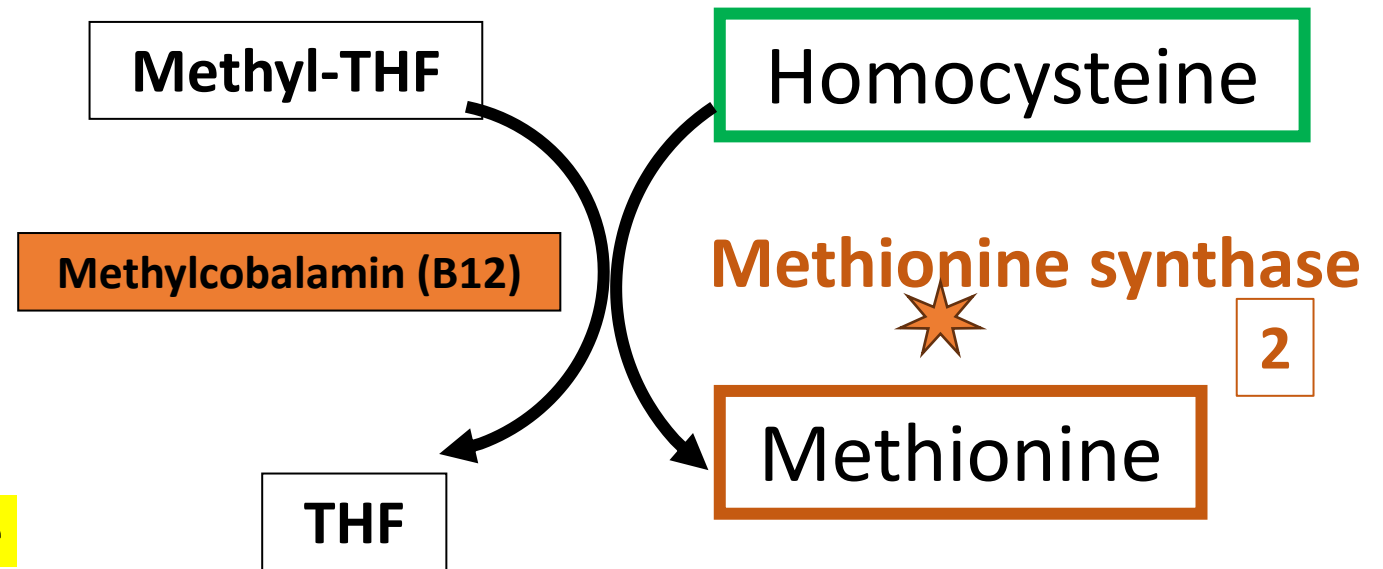
Amino acids: Isoleucine, valine
Odd-chain fatty acids



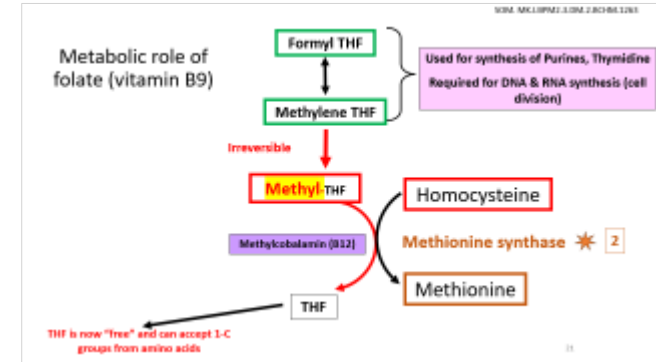
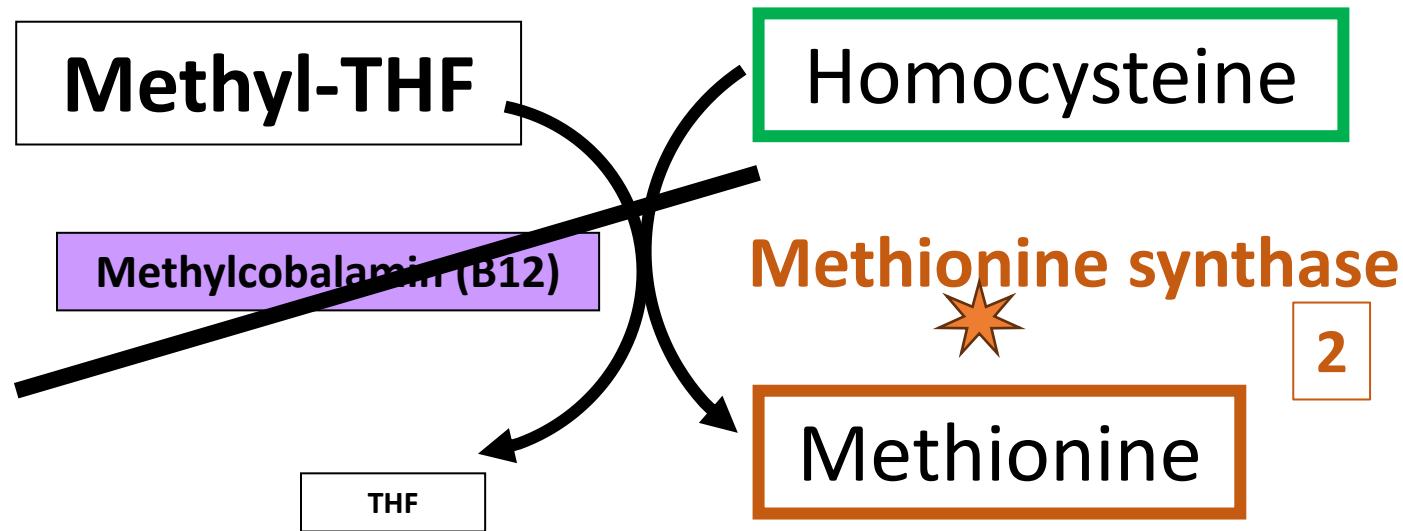
Reactions requiring vitamin B12

2. Methionine synthase (also requires methyl THF)

- Elevated serum **homocysteine** levels (In B₁₂ and folate deficiency and cannot distinguish folate from B₁₂ deficiency)
- Vitamin B₁₂ deficiency results in deficiency of usable form of folate (**Folate trap** hypothesis) → **Macrocytic anemia**
 - In B12 deficiency, folate levels might be normal or high;
 - But folate not usable because it is “trapped” in methyl-THF form (Secondary folate deficiency)



Vitamin B12 deficiency: Mechanism of **Folate trap** → Macrocytic anemia

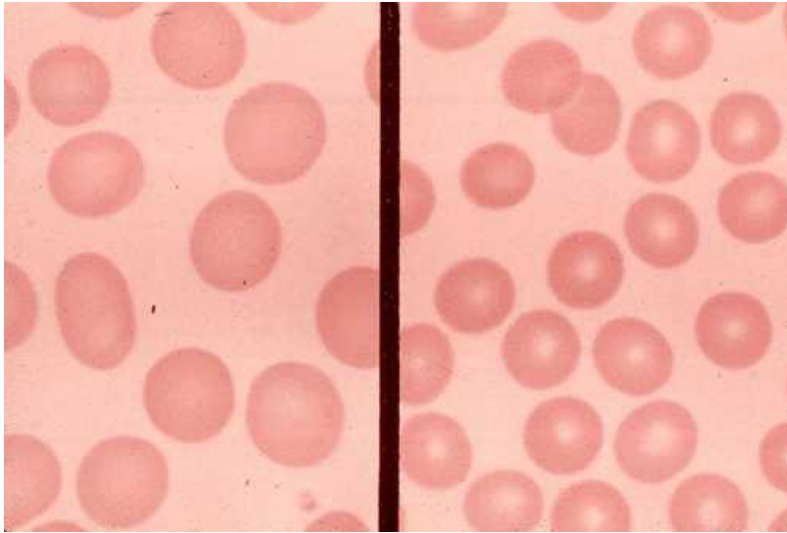


- Vitamin B₁₂ deficiency results in less availability of usable form of folate (**Folate trap hypothesis**) → **Macrocytic anemia**
- **Methyl-THF trap:** Methylene and formyl THF may eventually convert to methyl THF
- **Methyl-THF** cannot be converted to methylene and formyl-THF (irreversible)
- Methylene and formyl-THF become deficient → Reduced purine and thymidine synthesis → Reduced rates of DNA synthesis → Mitotic lag (**macrocytic anemia**)
- In B12 deficiency, folate levels might be normal or high; but “**trapped**” in **methyl-THF** form
- **Secondary folate deficiency/ functional folate deficiency**

Folic acid and vitamin B12 deficiency

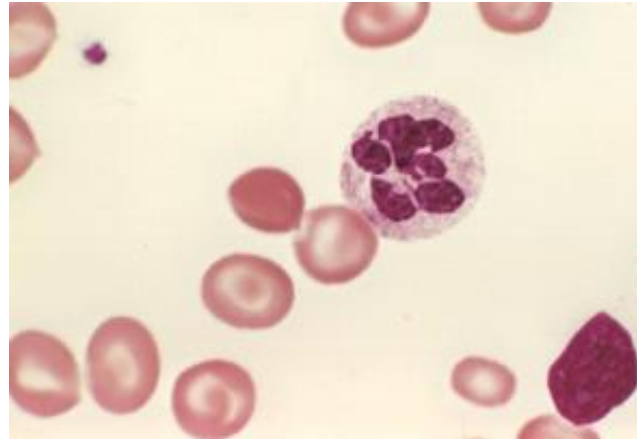
Macrocytic

Normal

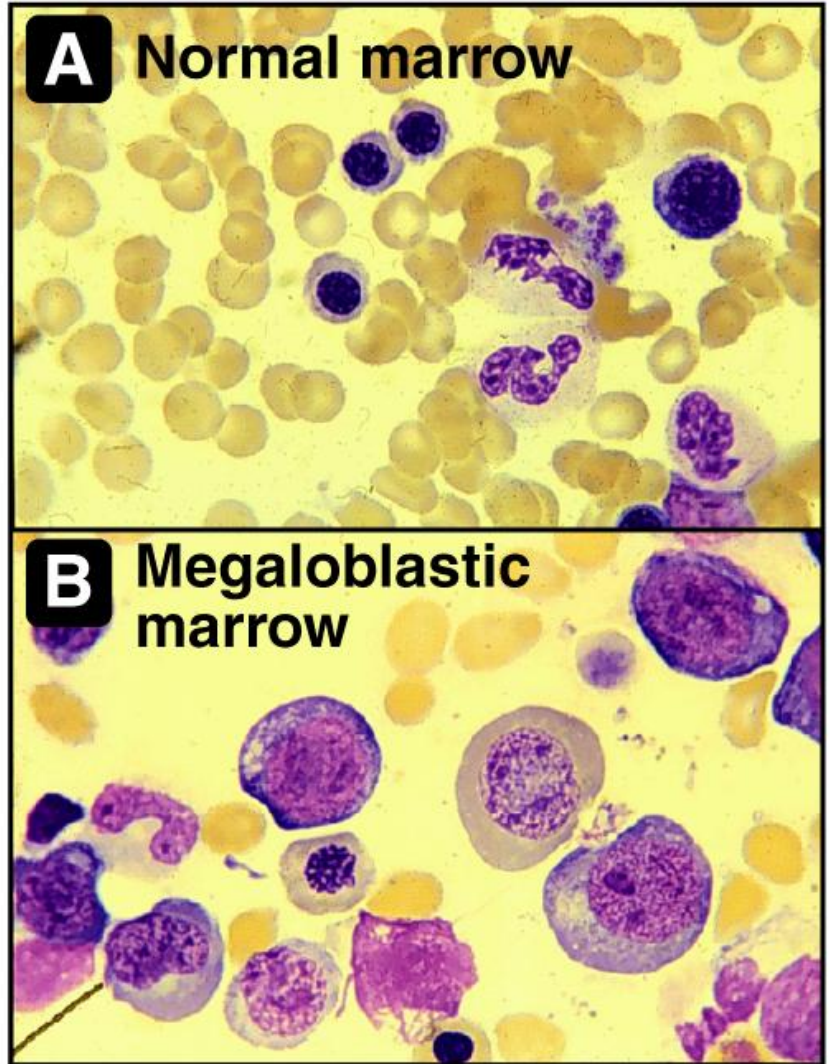


Mean
corpuscular
volume (MCV)
>100 fL ; Larger
RBC size

Normal MCV =
80-100 fL



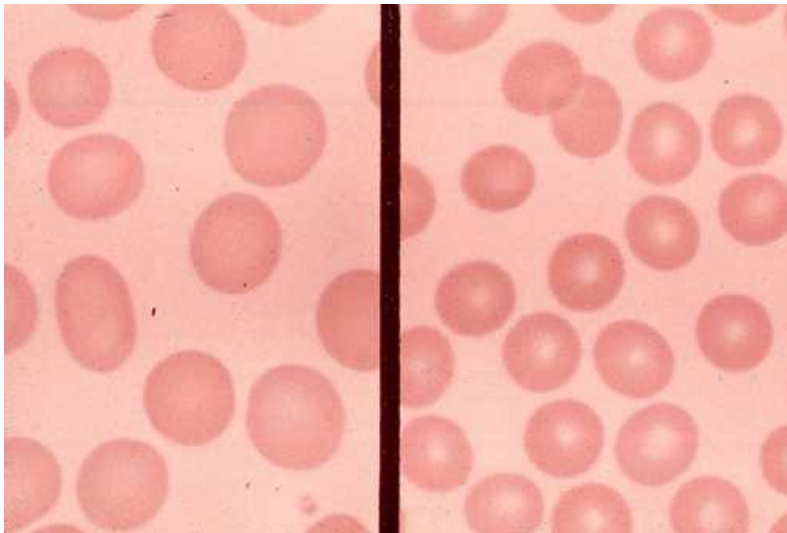
**Hypersegmented
neutrophils (>3-5
lobes)**



Macrocytic anemia: Folate and B12 deficiency

Macrocytic

Normal



**Mean corpuscular volume (MCV) >100 fL;
Larger average RBC size**

**Normal MCV =
80-100 fL**

**Folate and vitamin B12 deficiency
(Folate trap)**



**Reduced synthesis of Purines and
pyrimidine nucleotides**



**Reduced DNA synthesis (DNA
replication lags; S phase lag)**



**Cytosol enlarges but DNA synthesis
lags behind (MITOTIC lag)**



**Macrocytic RBC
precursors**

Case report: 1

- A 22-year-old woman in her first trimester of **pregnancy**, with pallor has hemoglobin concentration of **10 g/dl (Normal: 12-15)**.
- Lab values:
 - **Mean corpuscular volume (MCV) 121 fl** (normal 80-100);
 - **Serum folate 0.5 ng/ml** (normal >5.0 ng/ml);
 - Serum vitamin B₁₂ 1,500 pg/ml (normal, 200-800 pg/ml)
 - **Serum homocysteine: Elevated; Methylmalonate: Normal**
- She has peculiar food habits. She likes bread and occasionally chicken but diet has **no fresh vegetables** and rarely red meat or fish.

Folic acid deficiency during pregnancy

- Folate requirement increases during pregnancy
- Rapidly dividing fetal cells vulnerable to folate deficiency
- Higher risk of fetal **neural tube defects**
 - Spina bifida and anencephaly
 - Elevated maternal serum AFP levels (alpha-feto protein)
- Folate supplements pre-conception and during first trimester



Case report: 2

Beefy tongue



- A 14-year-old girl presents with complaints of **pallor, inability to walk, ataxia**, fever, and **fatigue** that had started a few months earlier and which had been getting worse in the past three weeks.
- Hemoglobin concentration is **8g/dL** (Normal: 12-15)
- Peripheral smear shows **macrocytic anemia** and **hypersegmented neutrophils**.
- **Serum B₁₂ level is 50** (normal: 160-970 pg/ml); Serum folate 8 ng/ml (normal>5.0 ng/ml)
- **Serum homocysteine and methylmalonate levels: Elevated**
- Her MRI findings show changes associated with **demyelination**.
- Diagnosis of tuberculosis of **terminal ileum** → Vitamin deficiency (Which vitamin and explain why?)
- After treatment of TB and vitamin supplements, she can eat, write, speak normally, walk, and ride a bicycle (neurological changes reversible in early stages).

Comparison of macrocytic anemias

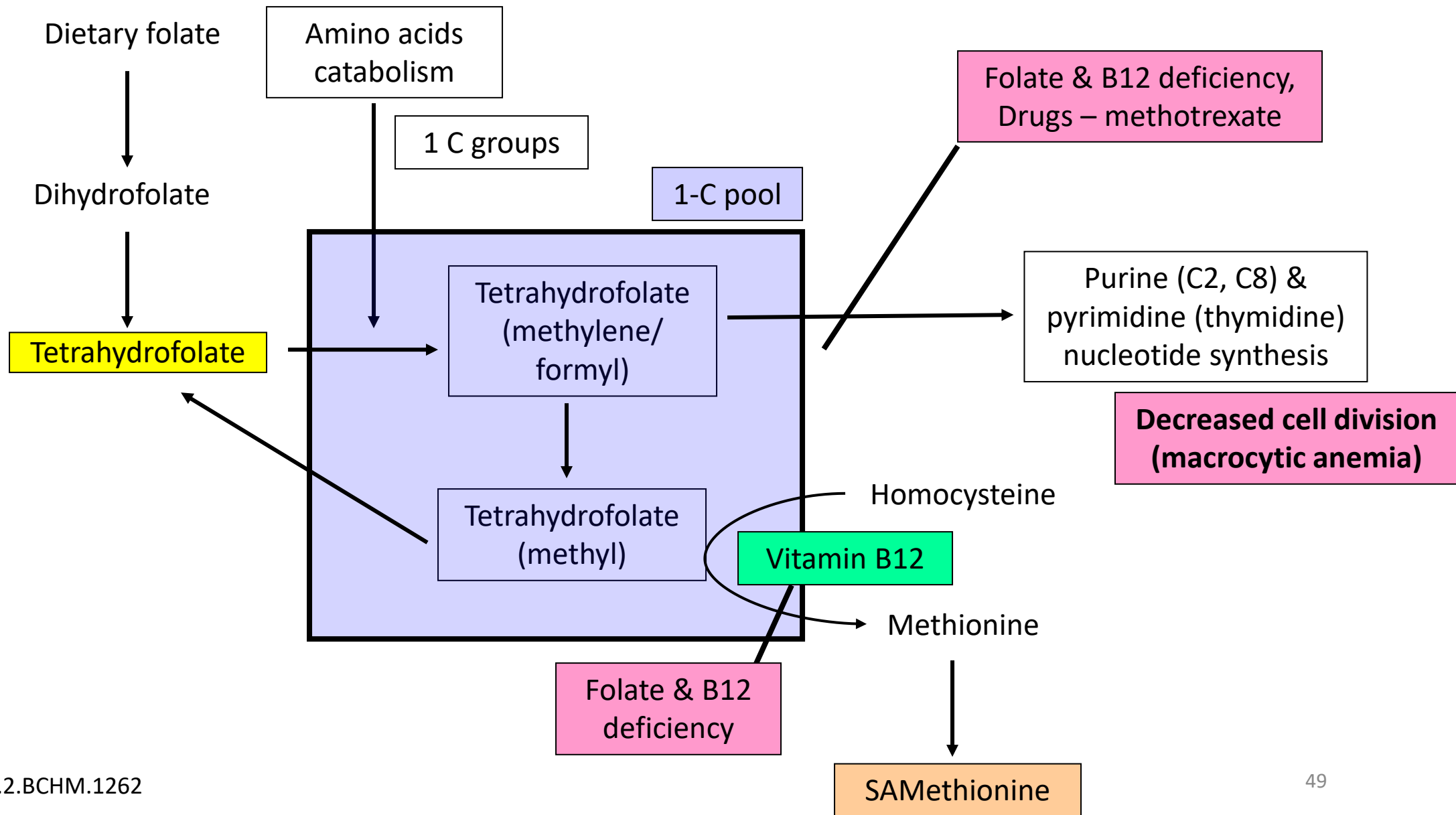
Vitamin B12 deficiency

- Macrocytic anemia with associated **neurological features**
- Anemia – Macrocytic (MCV increased) and megaloblastic bone marrow
- Serum **homocysteine** and serum **methylmalonate levels are elevated**
- Macrocytic anemia due to folate trap
- **Neurological features due to accumulation of methylmalonate (affects myelination)**
- Risk factors: Parietal cell destruction (pernicious anemia); ileal mucosal disease or resection; vegan diet
- Serum vitamin B12 levels are low
- Significant B12 liver stores

Folate deficiency

- Macrocytic anemia
- Anemia – Macrocytic (MCV increased) and megaloblastic bone marrow
- Elevated serum homocysteine levels; **Normal serum methylmalonate levels**
- Macrocytic anemia due to folate deficiency
- **NO neurological features**
- Risk factors: pregnancy, chronic alcoholics, folate antagonist treatment
- Serum folate levels are low
- No significant folate stores; Deficiency develops early

Summary of metabolic roles of vitamin B12 and folic acid



Thank you

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